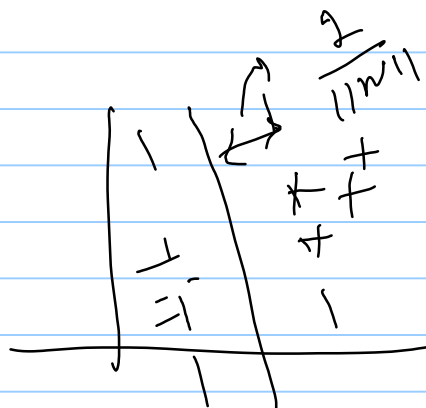


COL 774  
Machine Learning  
Sep 23, 2025



Convex  
General constrained min  
optimization:  $z$   
min  $f(w)$   
 $w$  — convex

$$g_i(w) \leq 0 \quad \forall i$$

$$h_\ell(w) = 0 \quad \forall \ell$$

2 affine  
(geometric  
linear)

$\Rightarrow$  special  
case

$$\frac{1}{2} w^T w + C \sum_i \epsilon_i$$

$$g_i(w) = \frac{1}{2} (w^T x_i + b) \geq 1 - \epsilon_i$$

$$\epsilon_i \geq 0$$

$$\Rightarrow L(w, b, \alpha, \gamma) = \frac{1}{2} w^T w + C \sum_i \epsilon_i$$

Lagrange  
multipliers

$$+ \sum_i \alpha_i [1 - g_i(w)]$$

$$+ \sum_i \gamma_i \epsilon_i$$

$$\mathcal{O}_p(w, b) = \max_{\alpha, \gamma} \quad \mathcal{L}(w, b, \alpha, \gamma)$$

$$\alpha, \gamma \geq 0$$

$\min_{w, b} (\mathcal{O}_p | w, b)$  solving  
 $\equiv$  original  
 problem

$$\mathcal{O}_D(\alpha, \gamma) = \min_{w, b} \quad \mathcal{L}(w, b, \alpha, \gamma)$$

$$\min_{w, b, \epsilon} \mathcal{O}_p(w, b, \epsilon) \geq \mathcal{O}_D(\alpha, \gamma)$$

$$\max_{\alpha, \gamma} \quad \alpha, \gamma \geq 0$$

Assume Convex Optimization problem  
 & Slater's condition

$$\begin{cases} \text{①} & \exists w^*, b^*, \epsilon^*, \alpha^*, \gamma^* \\ \text{②} & \mathcal{L}(w^*, b^*, \epsilon^*, \alpha^*, \gamma^*) = \mathcal{O}_p(w^*, b^*, \epsilon^*) = \mathcal{O}_D(\alpha^*, \gamma^*) \end{cases}$$

Primal Feasibility

Dual Feasibility

KKT:-

$$\text{①} \quad \epsilon_i \geq 0 \quad 1 - \epsilon_i - y_i(w_i z_i + b) \leq 0, \quad \alpha_i \geq 0, \gamma_i \geq 0$$

Derivative constraints

$$\text{②} \quad \nabla_{w, b, \epsilon} \mathcal{L}(w, b, \epsilon, \alpha, \gamma) \Big|_{w^*, b^*, \epsilon^*, \alpha^*, \gamma^*} = 0$$

complementary  
slackness

$$\textcircled{3} \quad \alpha_i^* g_i(w^*, b, \xi) = 0 \quad \forall i \quad \text{or} \quad \gamma_i^* \xi_i = 0$$

$$\Rightarrow \alpha_i^* > 0 \Rightarrow g_i(w^*, b, \xi) = 0$$

$$\gamma_i^* > 0 \Rightarrow \xi_i^* = 0$$

Dual:-

$$b = [1 - w^T x^u]_{y^u=1}$$

$$\alpha_i > 0, \gamma_i > 0$$

$$w = \sum_i \alpha_i x^u y^u$$

$$b = ? \quad \leftarrow \begin{matrix} \text{if} \\ \alpha_i > 0, \gamma_i > 0 \end{matrix}$$

$$\alpha_i + \gamma_i = C$$

SVM -  
Support Vectors:  $x^u$  s.t.  $\alpha_i > 0$

$$1 - \xi_i - y^u (w^T x^u + b) = 0 \Rightarrow b = 1 - w^T x^u$$

Dual:-

$$\max_{\alpha} \left[ \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j x^u T x^j y^u y^j \right]$$

$$\text{s.t. } \alpha_i \leq C$$

$$\sum_{i=1}^m \alpha_i y^u = 0$$

Dual  
problem